REQUIREMENT ANALYSIS

Technology Stack

Date	11 October 2026
Team ID	SWTID-2026-7837
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

1. Technology Overview

The project is implemented within the ServiceNow platform using its Service Catalog, Requested Items, Approval process, Flow Designer, and Catalog Tasks. Flow Designer is the core automation technology that connects an approved Standard Laptop request to automatic Catalog Task creation and assignment to the Hardware team.

2. Technology Stack

Technology / Component	Role in the Project	Project Usage
ServiceNow Platform	Enterprise workflow and service management platform	Hosts the Standard Laptop procurement workflow and related records.
Service Catalog	Request entry interface	Allows the user to submit the Standard Laptop request.
Requested Item (RITM)	Request context / procurement record	Represents the requested Standard Laptop item and provides context for the automation.
Approval Process	Approval control	Determines whether the Standard Laptop request is approved and can proceed to the automated task stage.
Flow Designer	Workflow automation	Detects the approved request and automates Catalog Task creation and assignment.
Catalog Task	Work item	Represents the hardware configuration task generated by the automation.
Hardware Team Assignment	Task routing	Routes the generated Catalog Task to the Hardware team for configuration.

3. Technology-to-Requirement Mapping

Requirement	Technology / Component	How It Supports the Requirement
Standard Laptop request	Service Catalog	Provides the entry point for the procurement request.
Approved request processing	Approval Process + Flow Designer	The approved outcome allows the automation to proceed.
Automatic Catalog Task creation	Flow Designer + Catalog Task	Creates the required task without manual task creation.
Hardware team assignment	Flow Designer + Task Assignment	Routes the generated task to the Hardware team.
End-to-end workflow	Service Catalog + Requested Item + Approval + Flow Designer + Catalog Task	Connects the procurement stages into a consistent automated process.
Improved task visibility	Catalog Task	Provides the Hardware team with a clear actionable work item.

4. Technology Architecture

Layer	Technology	Responsibility
Request Layer	Service Catalog	User submits the Standard Laptop request.
Record Layer	Requested Item	Stores the requested item context used by the workflow.
Approval Layer	ServiceNow Approval Process	Determines whether the request is approved.
Automation Layer	Flow Designer	Executes the automated approval-to-task workflow.
Task Layer	Catalog Task	Stores the actionable hardware configuration task.
Assignment Layer	Hardware Team	Receives and processes the generated task.

5. Technology Selection Rationale

• ServiceNow is used because the project is a ServiceNow-based IT procurement workflow.

• Service Catalog provides the required request entry point for the Standard Laptop order.

• Requested Items preserve the context of the requested procurement item.

• Flow Designer is used as the core automation mechanism to reduce manual task creation and assignment.

• Catalog Tasks provide a clear work item for the Hardware team.

• The selected components support a simple end-to-end workflow from request submission and approval to hardware configuration.

6. Development Environment

The solution is configured within a ServiceNow instance using the platform's Service Catalog, approval capabilities, Flow Designer, and Catalog Task functionality. The project does not specify any separate external programming language, database, or third-party service; therefore no additional technology is included in this document.

7. Technology Summary

The technology stack is focused on native ServiceNow workflow capabilities. Service Catalog and Requested Items capture the procurement request, the Approval process determines eligibility for automation, Flow Designer performs the workflow automation, and Catalog Tasks provide the Hardware team with the resulting configuration work item. Together, these components support the project's objective of reducing manual effort, delays, and errors in Standard Laptop procurement.